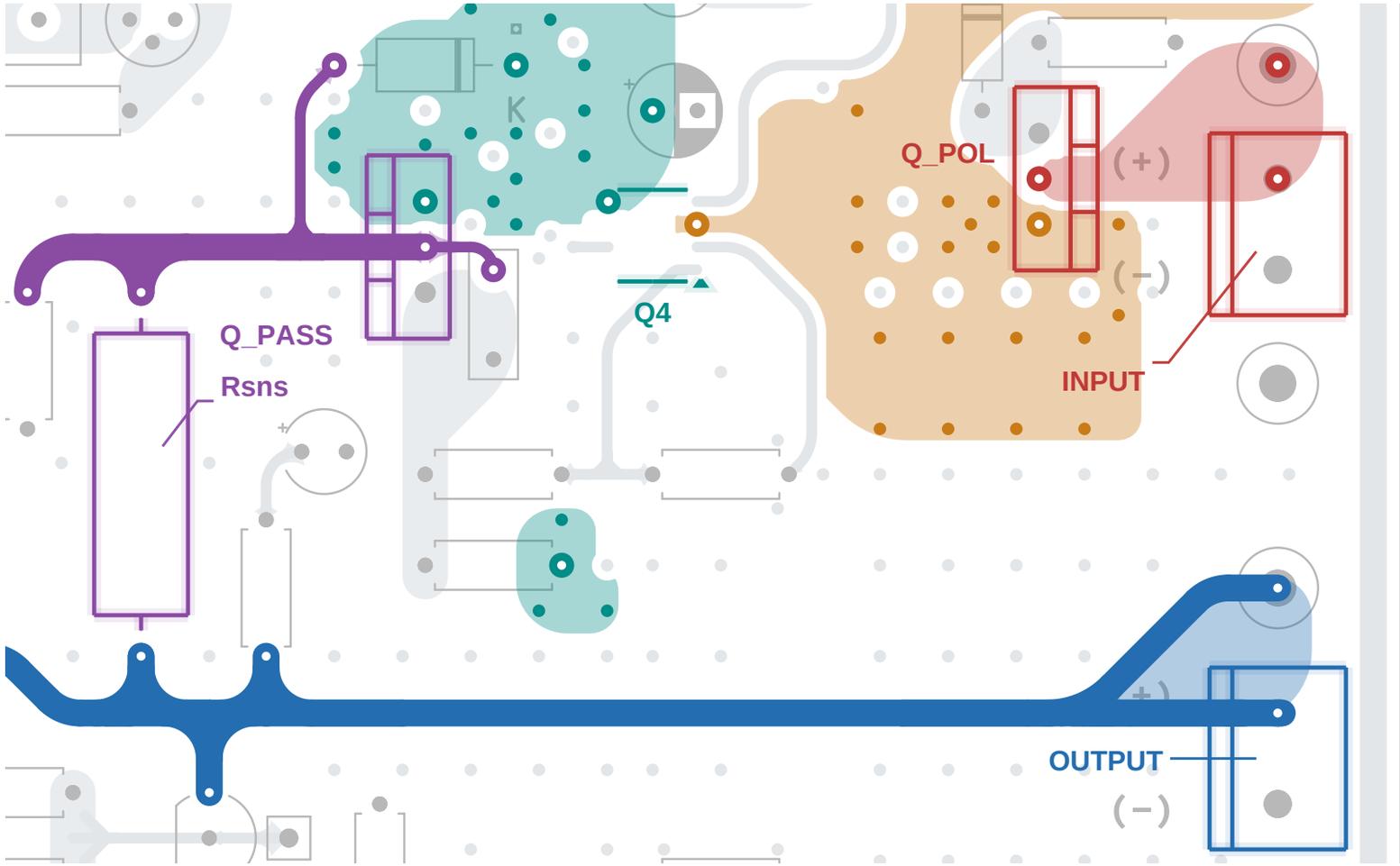
# Input-to-output power path

Actual KiCad copper + component highlights | Top copper and front silkscreen



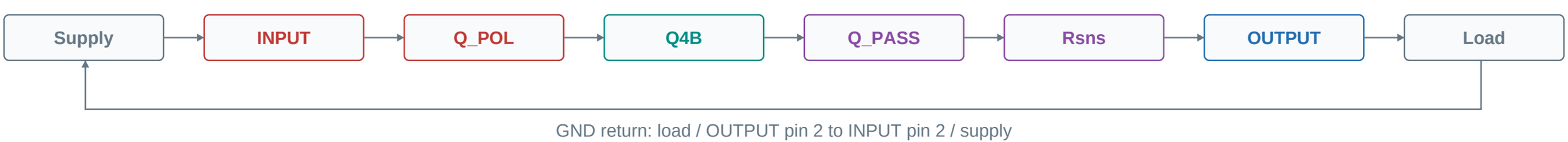

**Input and polarity protection**
VCC enters Q_POL at pin 2.

**Protected supply**
VEE joins Q_POL pin 3 to Q4B pin 3.

**Startup output**
Q4B pins 3-5 feed the pass emitter. Q4 is a dual-device package.

**Pass collector**
Q_PASS pin 2 reaches Rsns pin 1.

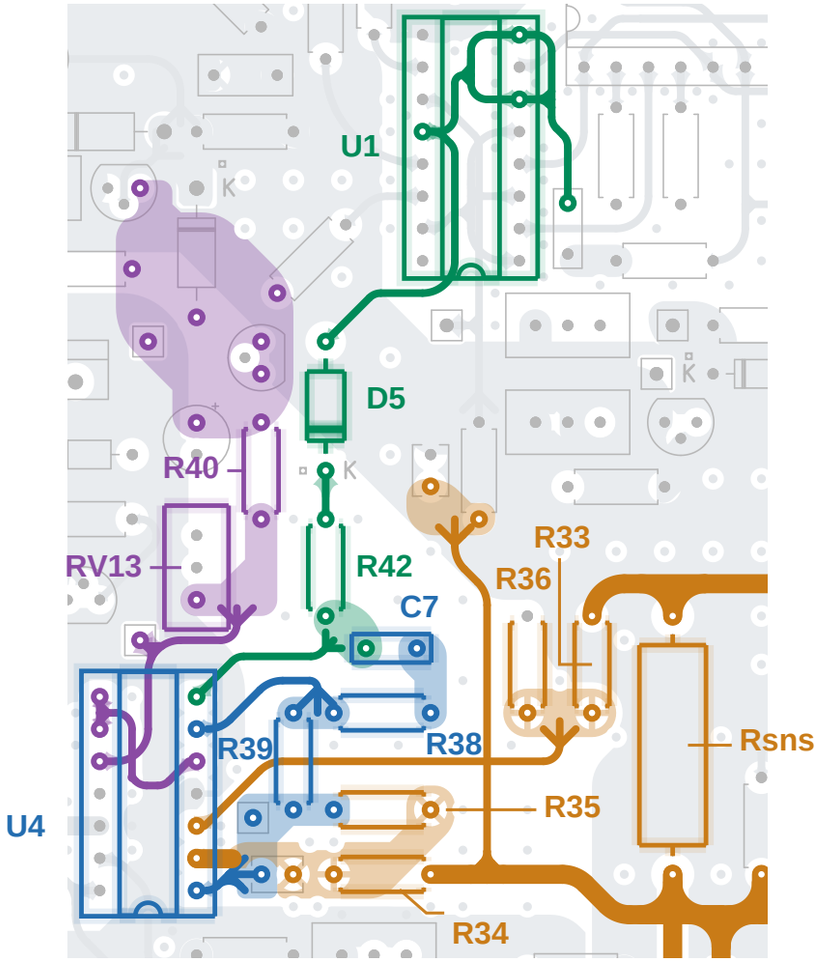
**Load output**
Rsns pin 2 reaches OUTPUT pin 1. The load is external.




Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

# Current sensing and limiting

Actual KiCad copper + component highlights | Top-side projection of both outer layers



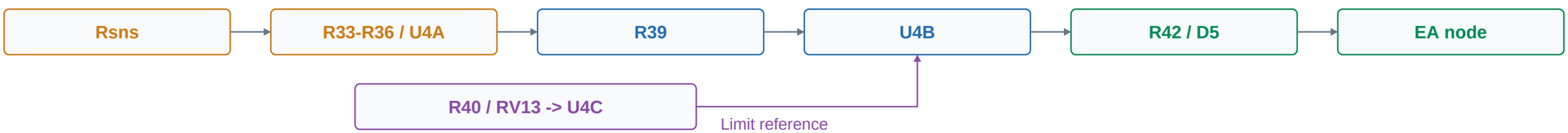

**Differential sensing**
R33 / R34 pick up opposite sides of Rsns. R35 / R36 complete the amplifier network.

**Amplified signal**
U4A output at pin 1 reaches U4B through R39. U4 contains several amplifier sections.

**Adjustable limit**
R40 / RV13 feed U4C pin 10. Its buffered threshold joins U4 pins 8, 9 and 5 via bottom copper and a via.

**Limiting action**
U4B output reaches the error-amplifier node through R42 and D5.

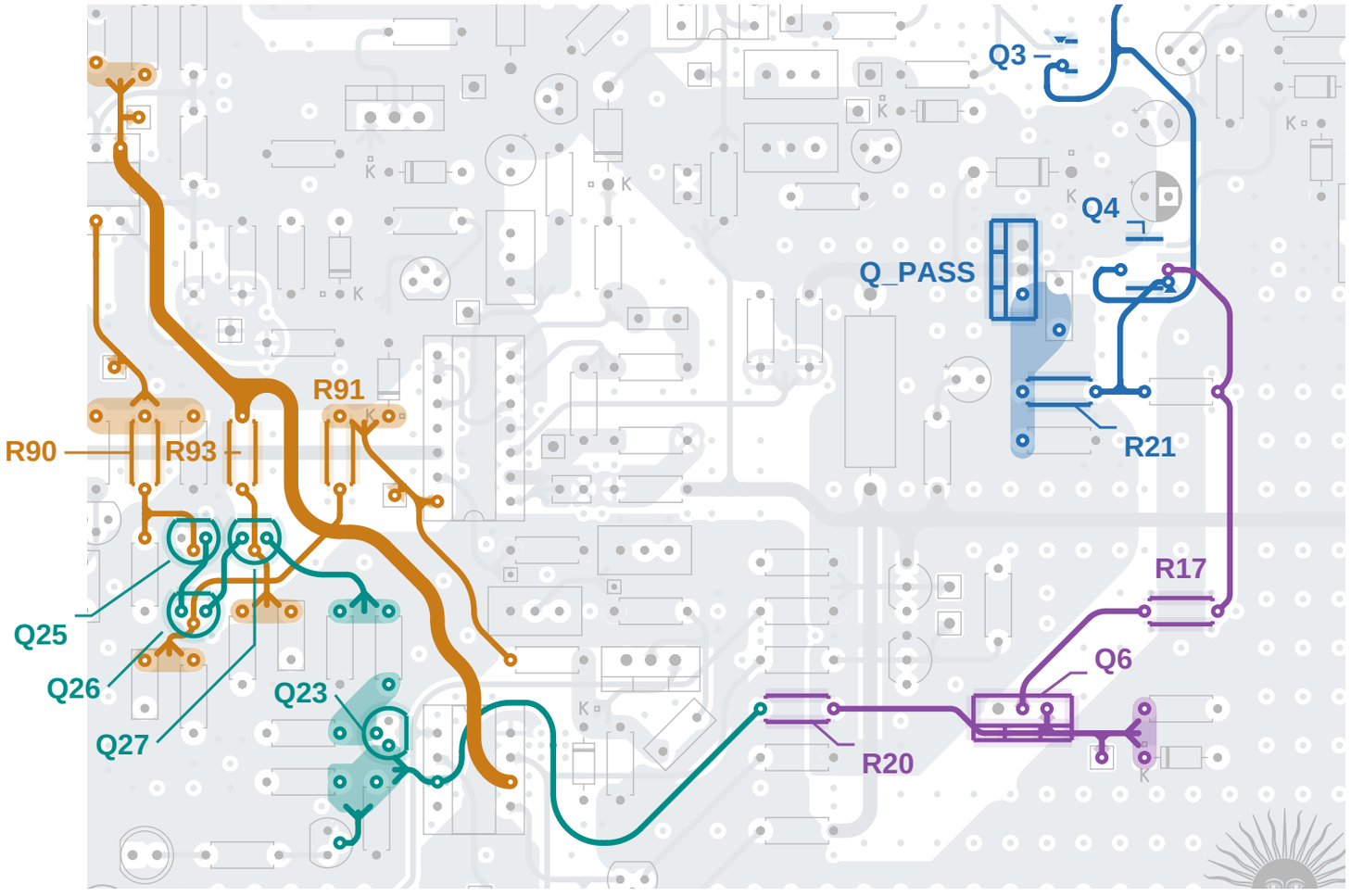
A shared color may span series components. Their pads and copper breaks remain distinct.




Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

# Enable logic and pass-drive switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers




Q4A and Q4B share the Q4 footprint. This view highlights the enable section Q4A.

**Permission inputs**

PCB nets 1 (UVLO), 2 (OVLO), and LDO2 (delay) drive the series NPN chain.

**Discrete AND function**

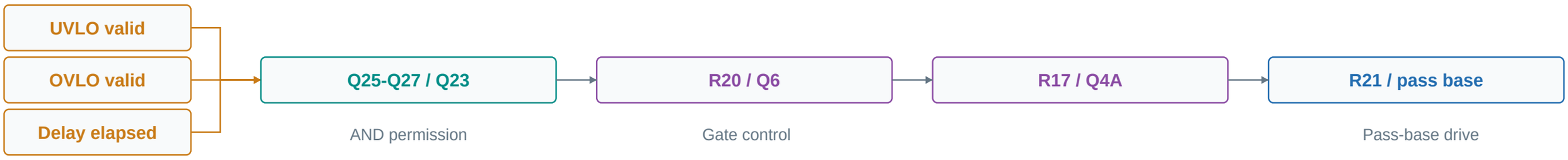
Q25-Q27 and the inverter Q23 generate the TTL enable node.

**Gate control**

TTL drives Q6 through R20. Q6 controls Q4A through R17.

**Pass-base drive**

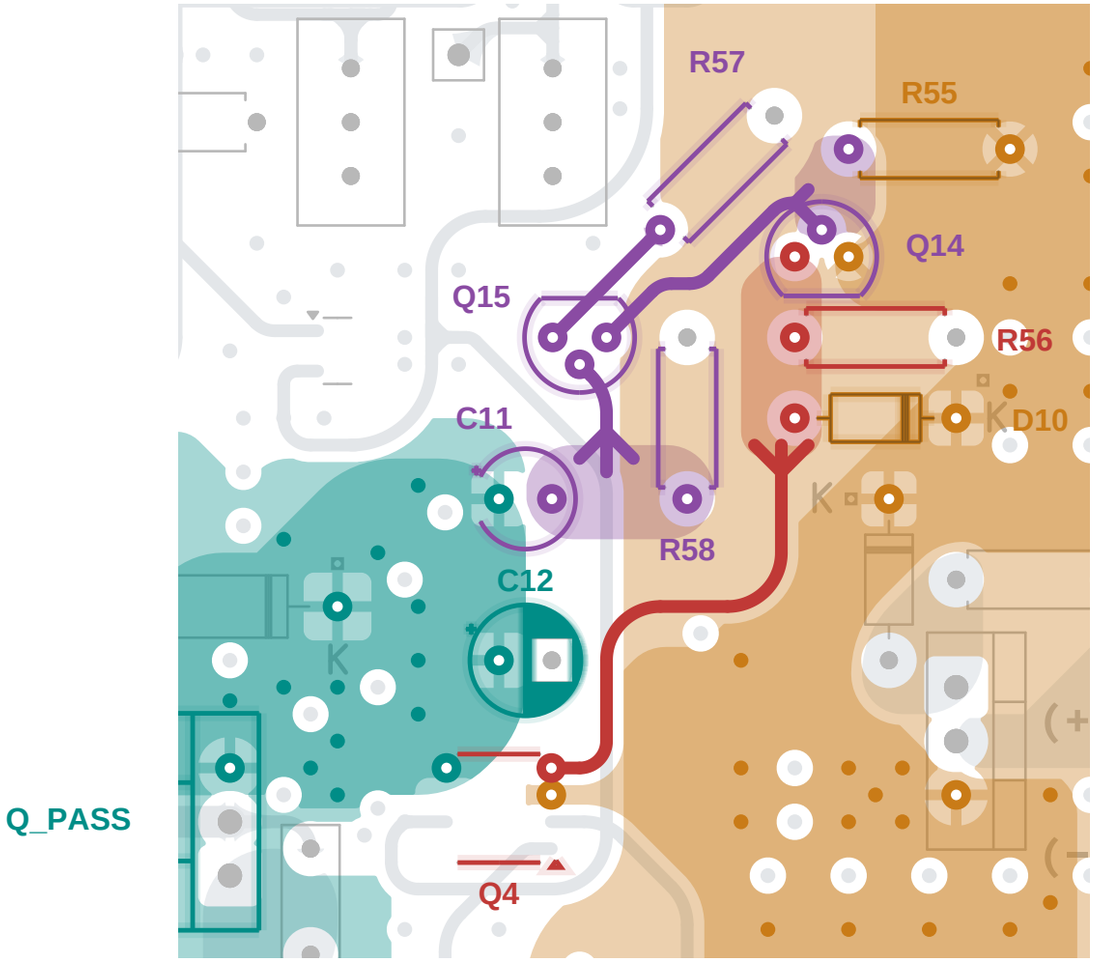
Q4A connects the Q3 output-stage path through R21 to the pass base.




Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.

# Startup ramp and series switch

Actual KiCad copper + component highlights | Top-side projection of both outer layers



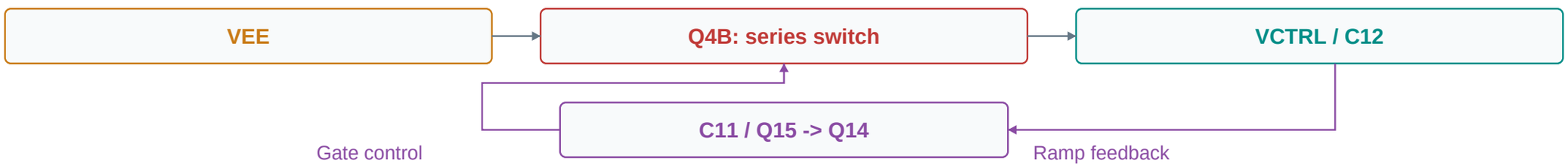

**Protected supply**
VEE supplies the startup stage and Q4B pin 3.

**Ramp feedback**
C11 couples changes of VCTRL into the Q15 / Q14 control network.

**Switch gate**
The Q14-controlled node drives Q4B gate pin 4; D10 connects it to VEE.

**Ramped supply**
Q4B pin 5 feeds VCTRL, C12 and the pass emitter. C12 is 10 uF in the PCB.

Q4B is the startup section of the shared Q4 package. The enable section Q4A is a separate path.




Ground fills omitted. Pad centers are schematic markers; colors group paths across series components. Not a fabrication drawing.